SYSE 2310: Systems Architecture

Instructor:	Aykut C. Satıcı	**Term:**	Fall 2026
Email:	aykut.satici@utdallas.edu	**Course Time:**	Tue & Thu 10:00 – 11:15 a.m.
Room:	ECSN 3.716	**Classroom:**	SLC 1.202

Course Page: https://elearning.utdallas.edu/ with Teams links.

Office Hours: Tuesdays and Thursdays 11:30 a.m. – 12:30 p.m. at ECSN 3.716, or by appt.

Prerequisites: This class requires an understanding and working knowledge of basic engineering principles. Exposure to basic probability and calculus is recommended.

Course Description: This course introduces systems architecture for complex engineered systems, emphasizing stakeholder needs, function-to-form mapping, concept generation, complexity management, and tradespace analysis. Students apply these methods in a semester-long Urban Air Mobility eVTOL project. LLMs support concept exploration and adversarial review, while students retain responsibility for all engineering models and conclusions.

Learning Outcomes:

1. *Analyze a complex system* by identifying its boundary, context, stakeholders, form, functions, relationships, interfaces, and emergent properties.
2. *Construct and interpret architectural representations* that decompose a system and trace externally delivered functions to internal functions and elements of form.
3. *Translate stakeholder needs into prioritized, measurable system goals* while accounting for strategic, regulatory, technological, operational, and lifecycle influences.
4. *Generate and compare solution-neutral system concepts* using functional decomposition, morphological exploration, and structured creativity.
5. *Develop and justify a coherent system architecture* by mapping function to form, defining interfaces and operational behavior, and managing complexity through modularization and decomposition.
6. *Formulate and evaluate architectural decisions* using value models, multi-attribute utility, Pareto analysis, and sensitivity analysis to identify preferred and robust alternatives.
7. *Critique architectural assumptions and recommendations*, including AI-generated content, by detecting unsupported claims, testing sensitivity to uncertainty, and documenting the engineering judgment behind final decisions.

Texts and Materials: The following textbook is required for this course. Classroom materials may not be reproduced or shared with those not in class. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

- E. Crawley, B. Cameron, and D. Selva, *Systems Architecture: Strategy and Product Development for Complex Systems*, Pearson, 2015.
- **AI Tools:** Students are expected to use UTD's freely available CometAI platform for AI assistance.

Optional/Supporting Resources: The following books will be helpful in understanding broader systems engineering concepts and methodologies:

- M. W. Maier and E. Rechtin, *The Art of Systems Architecting*, 3rd ed., Boca Raton, FL: CRC Press, 2009.
- D. M. Buede and W. D. Miller, *The Engineering Design of Systems: Models and Methods*, 3rd ed., Hoboken, NJ: John Wiley & Sons, 2016.
- S. Friedenthal, A. Moore, and R. Steiner, *A Practical Guide to SysML: The Systems Modeling Language*, 3rd ed., Waltham, MA: Morgan Kaufmann, 2014. (Optional deeper reference; SysML notation is not covered in class).
- K. T. Ulrich and S. D. Eppinger, *Product Design and Development*, 7th ed., New York, NY: McGraw-Hill Education, 2019.

Class Recordings: Students are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Unless the AccessAbility Resource Center has approved the student to record the instruction, students are expressly prohibited from recording any part of this course. Recordings may not be published, reproduced, or shared.

Grading Policy: Homework (25%), Midterm 1 (20%), Midterm 2 (20%), Project (35%).

- Homework must be submitted on the due date and time on eLearning. Extensions may be granted for special circumstances and only when requested at least one day in advance.
- You are responsible for all the information given in class verbally and/or in writing. Any information about the course on the web may be superseded by the information given in the class.
- **Project Milestones:** The 35% project grade is divided across four key milestones: Milestone 1: System Framing [AI-Assisted] (7%), Milestone 2: Form, Function, Architecture & Concept [AI-Assisted] (7%), Milestone 3: Stakeholders, Goals & Architecture [AI-Assisted] (10.5%), and Milestone 4: Value Model & Tradespace [Human-Authored, AI-Audited] (10.5%).

Homework Sequence (AI Literacy Arc): The homeworks build AI literacy intentionally and progressively, with each assignment noted as AI-supported or conventional when it is given. **AI-supported** assignments have the student use AI as a brainstorming and critique partner: one uses AI to generate alternative system-boundary and decomposition framings (graded on the judgment used to select and defend one), another uses AI to brainstorm morphological options (graded on pruning and filtering with engineering judgment), and another red-teams a student-built MAU model (graded on defending or correcting each AI critique). **Conventional** problem sets drill the underlying mathematical methods, establishing a baseline for independent technical reasoning.

Important Dates (Tentative):

Project Milestone 1	Thu, Sep 24
Midterm Exam 1	Thu, Oct 8
Project Milestone 2	Tue, Nov 3
Midterm Exam 2	Tue, Nov 17
Project Milestone 3	Thu, Nov 19
Project Milestone 4	Thu, Dec 10

Academic Honesty (Comet Creed): Academic dishonesty is **not** tolerated. Students are responsible for reading and understanding the student code of conduct. For more information, please see the student code of conduct.

Comet creed was voted by the UT Dallas student body in 2014. It is a standard that Comets choose to live by and encourage others to do the same:

As a Comet, I pledge honesty, integrity, and service in all that I do.

Accommodations for Students with Disabilities: Please review the section within the UT Dallas Syllabus Policies and Procedures webpage.

UT Dallas Syllabus Policies and Procedures: Please visit the Syllabus Policies page to view the University's policies and procedures segment of the course syllabus.

Tentative Course Outline:

Weeks 1 – 4: System Thinking and Complexity

- Crawley et al. Chapters 1 – 3
- Form vs. function, the Instrument-Process-Operand view, and emergence
- Decomposition and complexity heuristics
- In-class case study: the hospital emergency department as a complex system (feedback loops, self-organized criticality, edge of chaos), started in class and submitted as homework
- AI-assisted exploration of alternative system framings
- **Milestone 1 (System Framing; Chapter 2) due Thursday, September 24**

Weeks 5 – 8: Form, Function, Architecture, and Concept

- Crawley et al. Chapters 4 – 7
- Form and formal structure; function, internal functions, and the value pathway; architecture as the mapping of function to form
- Solution-neutral function, concepts, and morphological matrices
- In-class case study: a bicycle, analyzed one chapter at a time
- Counterbalanced concept-generation sprints (AI-assisted vs. unassisted) to compare design-space breadth
- **Midterm Exam 1 on Thursday, October 8**
- **Milestone 2 (Form, Function, Architecture & Concept; Chapters 4 – 7) due Tuesday, November 3**

Weeks 9 – 11: From Concept to Architecture, and Stakeholder Needs

- Crawley et al. Chapters 8 – 11
- From concept to architecture: Level 2 decomposition and modularization
- The architect's role: resolving ambiguity
- Upstream/downstream influences; translating stakeholder needs into prioritized goals
- AI stakeholder personas: eliciting needs and detecting hallucinations
- **Milestone 3 (Stakeholders, Goals & Architecture; Chapters 8 – 11) due Thursday, November 19**

Weeks 12 – 16: Decisions, Value Modeling, and Tradespace Exploration

- Crawley et al. Chapters 12 – 16
- Creativity in concept refinement; managing complexity
- Architecture as a decision-making process
- Value models: utilities and weights (authored independently)
- Tradespaces, Pareto frontiers, and sensitivity analysis (authored independently)
- AI audit of the student's own tradespace analysis
- **Midterm Exam 2 on Tuesday, November 17**
- **Milestone 4 (Value Model & Tradespace; Chapters 14 – 15) due Thursday, December 10 (last day of classes)**